Yatharth Goyal

AI / ML | MODEL EVALUATION · COMPUTER VISION · OPTIMIZATION

Jaipur, India | yatharthgoyal657@gmail.com
GitHub / LinkedIn / Java DSA / Portfolio source

TECHNICAL SKILLS

Languages Python, Java, SQL, JavaScript
ML and data scikit-learn, YOLO, pandas, NumPy, SciPy/HiGHS, MILP
Systems FastAPI, Express, PostgreSQL/PostGIS, Redis/BullMQ, React
Validation pytest, Git, Docker; data audits, model evaluation; Java DSA

SELECTED PROJECTS & ENGINEERING CONTRIBUTIONS

AegisED

SYNTHETIC ML · PUBLIC SOURCE

- Integrated a FastAPI/scikit-learn triage-support service with human review and recorded overrides in an emergency-operations platform.
- Documented macro-F1 of 0.7893 on 900 held-out synthetic cases; this is non-clinical evaluation, not clinical validation.

RouteGuard 2.0

IN PROGRESS

- Evaluating YOLO road-damage models through label reviews, dataset-integrity checks and split-leakage audits; integrating PostGIS-backed route comparison.
- Research target: 95%+ precision AND recall per severe-hazard class on independent test data. Target not achieved.

EvalForge

PRIVATE · RELEASE PENDING

- Built rubric-grounded evaluation with exact response spans, output validation and instructor overrides; separated workers from durable PostgreSQL jobs.
- Documented 7/7 evaluator and 15/15 backend local checks; no externally validated scoring accuracy or public deployment is claimed.

Grid Flex

LOCAL PROTOTYPE

- Formulated synthetic energy scheduling as a SciPy/HiGHS MILP with cost, balanced and carbon objectives; added independent constraint audits.
- Documented 52/52 local tests and scenario stress analysis; results are modeled, not measured industrial savings. AWS SAM is not deployed.

EDUCATION & CERTIFICATIONS

JECRC | B.Tech Information Technology
Expected graduation: 2028 | CGPA: 9.00/10.00